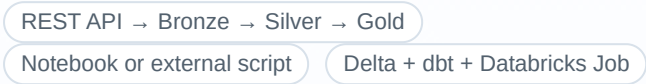


Databricks Execution Flow

Repository: Girish-LS/senior-de-assignment

End-to-end execution of the Databricks path for transaction ingestion, medallion-layer transformation, orchestration, and idempotent reruns.



Verified repository evidence

- First bronze run: **352 fetched, 349 valid, 3 quarantined, 5 duplicate rows flagged across 5 duplicate groups**
- Incremental rerun: **17 re-read**, bronze row count stays **349**
- dbt on Databricks: **PASS=34 WARN=0 ERROR=0**
- Output artefacts committed under `outputs/databricks/`

Bronze

Raw STRING-preserving Delta layer plus quarantine, watermark, and metrics.

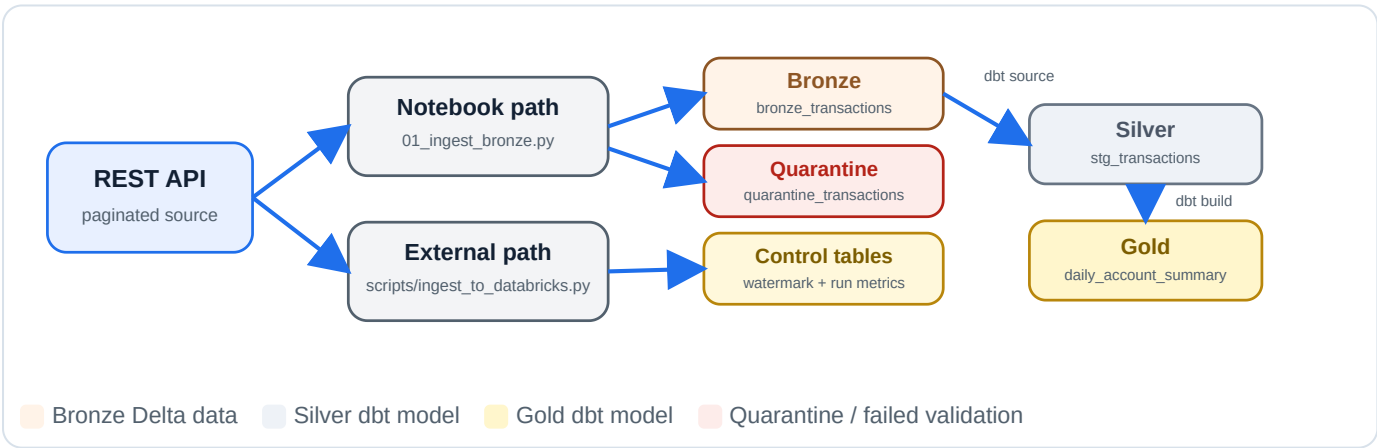
Silver

stg_transactions deduplicates survivors and casts types once.

Gold

daily_account_summary incrementally recomputes account/day aggregates.

1. High-level architecture



2. Databricks object map

Layer	Object	Contents
Bronze	workspace.bronze.bronze_transactions	Validated source records persisted as Delta with every business field stored as STRING plus natural_key_hash, duplicate flags, and ingestion metadata.
Bronze	workspace.bronze.quarantine_transactions	Records that failed validation, including full raw payload, all failure reasons, error count, and failed fields.
Bronze	workspace.bronze.pipeline_watermark	Single-row high-water mark per source_system, computed from valid records only and written after a successful run.
Bronze	workspace.bronze.ingestion_run_metrics	One row per run: mode, status, fetched/valid/quarantined/duplicate counts, watermark before/after, duration, request stats.
Silver	workspace.silver.stg_transactions	Deduplicated survivor rows with typed amount and transaction_day; the single definition of a usable transaction.
Gold	workspace.gold.daily_account_summary	One row per account per UTC day with debit, credit, net, count, distinct merchants, top category, currencies, and update timestamp.

3. Two execution paths write the same bronze layer

Notebook path (production-oriented)

File:

databricks/notebooks/01_ingest_bronze.py

- Runs on Databricks serverless compute.
- Reads credentials from secret scope transactions-api.
- Parameters: mode, catalog, bronze_schema, secret_scope, lookback_hours.
- Imports the repository's own validation and dedupe code instead of reimplementing rules in the notebook.

```
dbutils.widgets.dropdown("mode",
"incremental", ["incremental", "full"])
dbutils.widgets.text("catalog", "workspace")
dbutils.widgets.text("bronze_schema",
"bronze")
dbutils.widgets.text("secret_scope",
"transactions-api")
dbutils.widgets.text("lookback_hours", "72")
```

External script path (workspace-light)

File:

scripts/ingest_to_databricks.py

- CLI execution from a workstation or automation shell.
- Uses databricks-sql-connector.
- Reads DATABRICKS_HOST, DATABRICKS_HTTP_PATH, DATABRICKS_TOKEN, and optional catalog/schema overrides.
- Writes the same four bronze objects and reuses the same pipeline components.

```
python
scripts/ingest_to_databricks.py --
mode full
python
scripts/ingest_to_databricks.py
# incremental
```

Both paths converge on the same storage design: Delta bronze tables in Unity Catalog followed by dbt-driven silver and gold models.

4. Step-by-step bronze execution flow

1. **Connect and create objects idempotently.** The Databricks path creates schema bronze and four Delta tables if they do not already exist.
2. **Decide full vs incremental mode.** Full load reads everything. Incremental mode reads the stored watermark and applies a 72-hour trailing lookback window.
3. **Fetch from the REST API.** Both Databricks endpoints use `TransactionsApiClient.iter_transactions()`.
4. **Validate each record.** Records either become typed `Transaction` objects or accumulate all validation failures.
5. **Flag duplicates.** Duplicate detection hashes all business fields except `transaction_id`; the lowest `transaction_id` survives.
6. **Write quarantine rows.** Failed records land in `quarantine_transactions`; a run clears its own quarantine rows first to stay idempotent within that run.
7. **Upsert bronze via MERGE.** Valid records are staged and merged into `bronze_transactions` on `transaction_id`.
8. **Advance watermark only after success.** The new watermark is the max valid `transaction_date` if it is higher than the previous value.
9. **Record metrics.** Each run writes a metrics row capturing counts, timing, and watermark movement.

Watermark management

Why gte instead of gt? The API filter intentionally re-reads the boundary timestamp so records that share the exact watermark are not skipped forever.

```
effective_from = apply_lookback(watermark_before, 72)
# filter passed to the API: gte.<effective_from>
```

Why compute the watermark from valid records only? This dataset contains invalid records dated 2024-04-15 and 2024-11-31, later than every valid record. A raw-record watermark would jump to November and silently stop future ingestion.

Zero-record guard. The current checked-in Databricks endpoints — `databricks/notebooks/01_ingest_bronze.py` and `scripts/ingest_to_databricks.py` — each include a guard that raises instead of reporting success when an incremental run with a lookback window fetches zero rows, because that window should still contain the watermark-setting record. The code comments tie this to the historical malformed-filter case `gte.gte.<timestamp>`.

Validation and quarantine

- Validation enforces strict UTC timestamp format, positive decimal amount, accepted enumerations, non-blank merchant names, and assigned ISO 3166-1 alpha-2 country codes.
- Every failure reason is retained; validation does not stop at the first error.
- Quarantine rows store the raw JSON record, a semicolon-separated error message, error count, and failed field list.

```
quarantine_id, transaction_id, raw_record, error_reason, error_count,
failed_fields, ingestion_timestamp, ingestion_run_id, source_system
```

Duplicate handling

- Duplicates are **flagged in bronze**, not dropped at ingestion time.
- The survivor is deterministic: the lowest `transaction_id` in each duplicate group.
- Silver resolves the correctness choice by filtering `is_duplicate = 0`.
- Committed evidence shows **5 duplicate groups** under `outputs/databricks/duplicate_groups.csv`.

MERGE INTO upsert pattern

```

MERGE INTO workspace.bronze.bronze_transactions AS target
USING bronze_incoming AS source
  ON target.transaction_id = source.transaction_id
WHEN MATCHED      THEN UPDATE SET *
WHEN NOT MATCHED THEN INSERT *

```

This is the Delta equivalent of the local SQLite `INSERT ... ON CONFLICT` pattern. It is the mechanism that makes the deliberate gte overlap safe.

5. dbt transformation path

Silver: stg_transactions

- Materialized as a **view**.
- Reads Databricks bronze as a dbt source whose schema is explicitly set to bronze.
- Casts amount once at the silver boundary.
- Creates `transaction_day` from the timestamp string.
- Filters out duplicate rows with `where is_duplicate = 0`.

```

select *
from {{ source('raw',
'bronze_transactions') }}
where is_duplicate = 0

```

Gold: daily_account_summary

As currently defined in `dbt_project/models/marts/daily_account_summary.sql` and `dbt_project/models/marts/schema.yml`:

- `materialized='incremental'`, `unique_key=['account_id', 'transaction_date']`, `incremental_strategy='delete+insert'`, and `on_schema_change='fail'`.
- Includes only rows where `status = 'completed'`.
- Incremental predicate recomputes days where `transaction_day >= max(transaction_date) - interval '{{ var(\"summary_lookback_days\") }} days'`.
- The model contract is enforced in `dbt_project/models/marts/schema.yml`.

```

{{ config(
  materialized='incremental',
  unique_key=['account_id',
'transaction_date'],
  incremental_strategy='delete+insert'
) }}

```

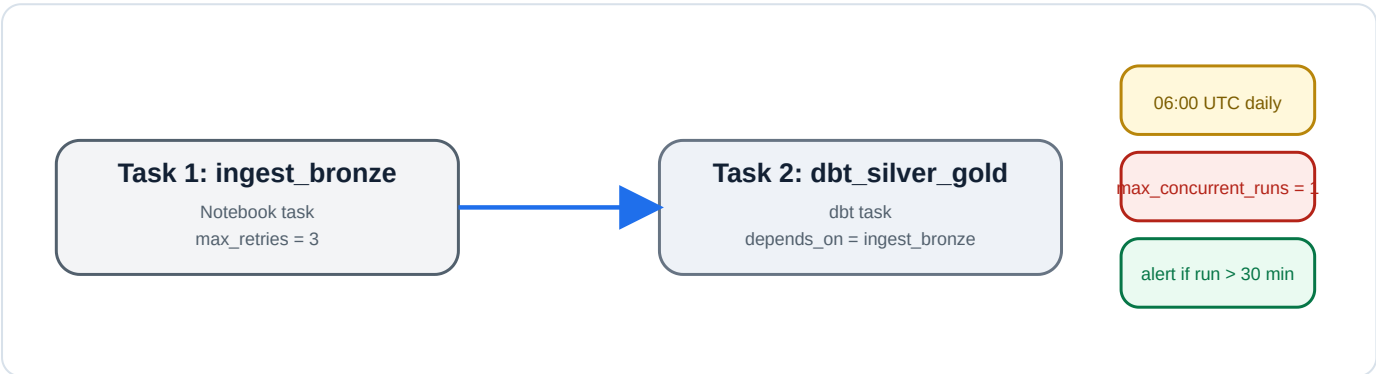
The exact Databricks build command is:

```
cd dbt_project
dbt deps
dbt build --target databricks
```

Committed output in outputs/databricks/dbt_build.txt shows **2 models, 32 data tests, 1 exposure**, and final status **PASS=34 WARN=0 ERROR=0**.

6. Orchestration job definition and task dependencies

Diagram values below are taken from the current checked-in job definition:
databricks/jobs/transactions_pipeline_job.json.



Task	Type	Retries	Depends on	Reasoning
ingest_bronze	Notebook task	3, 60s interval	—	API/network failures are often transient and should not force a full pipeline rerun.
dbt_silver_gold	dbt task	1	ingest_bronze	Failing data tests are usually not transient; retrying fetch plus transformation together would be wasteful and misleading.

7. Portability findings from the DuckDB-first dbt path to Databricks

Finding	What worked locally	Databricks outcome / resolution
timestamp with time zone	DuckDB contract accepted a timezone-aware timestamp type.	Spark SQL does not expose that type in the same way. Resolved by normalizing the model output to plain timestamp.
VARCHAR in the dbt contract/cast path	DuckDB accepted the generated <code>cast(null as varchar)</code> shape used by the model contract.	That specific Databricks/Spark path raised <code>DATATYPE_MISSING_SIZE</code> ; the repo resolved it by declaring <code>string</code> , which Databricks accepts and DuckDB treats as an alias.
<code>listagg order_by</code>	DuckDB honors ordering for the currencies list.	Spark/Databricks ignores <code>order_by_clause</code> in dbt's <code>listagg</code> . This remains documented rather than hidden.
UTF-8 BOM	DuckDB tolerated files that started with a BOM.	Spark SQL parsing failed; the repo explicitly documents the issue and already handles BOMs in config file reading.

The broader conclusion is that dbt model contracts are expressed in the destination warehouse's type system. Cross-engine portability therefore requires explicit output normalization, not just SQL that “looks portable”.

8. Key design decisions

- gte instead of gt:** prevents permanent loss of records that share the exact watermark timestamp.
- Watermark from valid rows only:** avoids the invalid-date trap created by later-dated quarantined records.
- Zero-record guard:** catches malformed incremental filters rather than treating an impossible empty window as success.
- Duplicates flagged in bronze, resolved in silver:** keeps source-quality observability while making the correctness choice reversible downstream.
- Bronze stores raw values as STRING:** preserves source fidelity and keeps money out of floating-point representation before silver decides types.
- Gold uses `delete+insert`, not `merge`:** aggregate rows must be recomputed from all contributing records when late-arriving data lands.

9. Committed evidence in outputs/databricks/

File	What it proves
run_transcript.txt	Full end-to-end execution transcript for full ingestion, incremental rerun, and dbt build.
ingest_run1.json	Initial full run: watermark was null, 352 fetched, 349 valid, 3 quarantined, and 5 duplicate rows flagged across 5 duplicate groups.
ingest_run2.json	Incremental rerun: watermark before and after both 2024-03-30T21:01:36Z, 17 rows re-read from the lookback window.
bronze_sample.csv	Sample raw-plus-metadata bronze rows written to Delta.
quarantine_sample.csv	Invalid records with all reasons preserved.
duplicate_groups.csv	Five duplicate groups and their deterministic survivors.
daily_account_summary.csv	Gold output produced from the Databricks path.
watermark.csv	Current watermark state in the bronze control table.
run_metrics.csv	Per-run operational history.
idempotency_check.txt	Explicit bronze count before vs after rerun: 349 → 349.
table_counts.json	Compact summary of table counts and current watermark.
dbt_build.txt	Databricks dbt success output: PASS=34 WARN=0 ERROR=0.

10. Idempotency proof

Evidence from committed artefacts:

- ingest_run1.json: watermark_after = 2024-03-30T21:01:36Z
- ingest_run2.json: same watermark_before and watermark_after, with effective_filter_from = 2024-03-27T21:01:36Z
- idempotency_check.txt: bronze row count remains **349** before and after the rerun

```
bronze rows before rerun : 349
bronze rows after rerun  : 349
result                   : PASS - zero new rows
```

Why the proof holds:

1. The incremental filter deliberately re-reads a trailing window using gte.
2. MERGE INTO matches on transaction_id and updates existing bronze rows instead of duplicating them.
3. The watermark advances only after a successful write and only to the maximum valid transaction date.

The combination of those three choices makes incremental overlap safe, measurable, and repeatable.

11. Execution summary

The Databricks path is a medallion-style pipeline with two interchangeable ingestion entrypoints, a shared validation/deduplication core, Delta bronze control tables, dbt-modeled silver and gold layers, and a job

definition that keeps ingestion and transformation independently retryable. The design choices around gte, valid-only watermarking, and Delta MERGE INTO are not isolated implementation details; together they are the idempotency strategy.